# Topic 20: Organic Synthesis

## Students will be assessed on their ability to:

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| **20.1** | be able to deduce the empirical formulae, molecular formulae and structural formulae from data drawn from combustion analysis, element percentage composition, characteristic reactions of functional groups, infrared spectra, mass spectra and NMR spectra (both $^{13}C$ and proton) |
| **20.2** | understand methods of increasing the length of the carbon chain in a molecule by the use of magnesium to form Grignard reagents and the reactions of the latter with carbon dioxide and with carbonyl compounds in dry ether |
| **20.3** | be able to use knowledge of organic chemistry contained given in this specification to solve problems such as:<br>i predicting the properties of unfamiliar compounds containing one or more of the functional groups included in the specification and explain these predictions<br>ii planning reaction schemes of up to four steps, recalling familiar reactions and using unfamiliar reactions given sufficient information<br>iii selecting suitable practical procedures for carrying out reactions involving compounds with functional groups included in this specification<br>iv identifying appropriate control measures to reduce risk based on data of hazards |
| **20.4** | **CORE PRACTICAL 16**<br>**The preparation of aspirin.** |
| **20.5** | understand the following techniques used in the preparation and purification of organic compounds:<br>i refluxing<br>ii purification by washing, including with water and sodium carbonate solution<br>iii solvent extraction<br>iv recrystallisation<br>v drying<br>vi distillation<br>vii steam distillation<br>viii melting temperature determination<br>ix boiling temperature determination |
| | **Further suggested practicals:**<br>i carry out the preparation of an organic compound, including cholesteryl benzoate (a liquid crystal) or methyl 3-nitrobenzoate<br>ii preparation of oil of wintergreen |